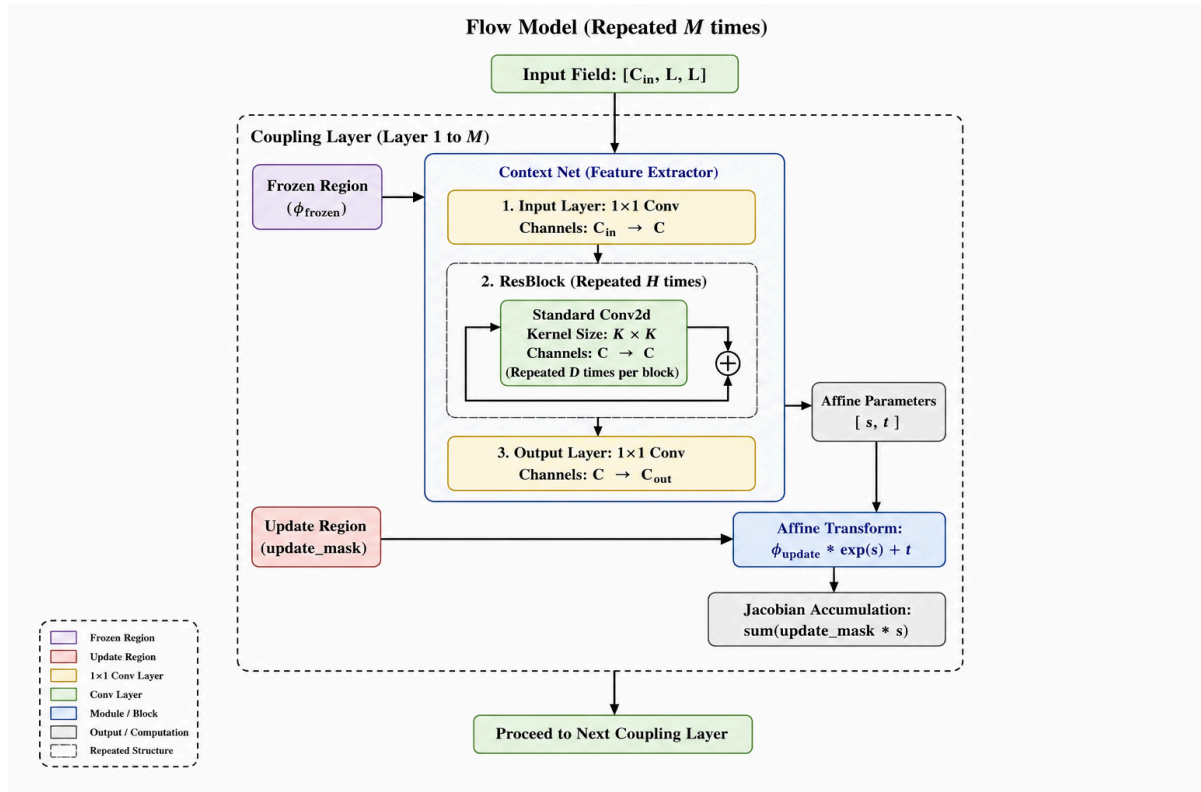


FLOPs Analysis And Autocorrelation Time

1. Architecture Overview

This network is a Normalizing Flow model designed for scalar field configuration generation. It utilizes purely local convolutional operations to achieve UV-IR decoupling.

A single forward mapping step consists of M stacked **Coupling Layers**. Each Coupling Layer partitions the spatial lattice into two regions using a checkerboard mask. It freezes one half of the physical field (ϕ_{frozen}) and passes it through a **Context Network** (a feature extractor). The Context Network is a deep convolutional architecture built from standard Residual Blocks (ResBlocks). It outputs the affine transformation parameters (scale s and translation t) which are then used to update the unmasked half of the field.



2. Abstract FLOPs Calculation Formula

To calculate the Floating Point Operations (FLOPs), we define the following variables corresponding to the architecture:

- L : **Lattice Size**. Spatial resolution of the field ($L \times L$).
- C_{in} : **Input Channels**. Number of physical field channels.
- C_{out} : **Output Channels**. Number of affine parameter channels.
- M : **Coupling Layers**. Total number of sequential Coupling Layers.
- C : **Hidden Channels**. Internal channel width used throughout the Context Net.
- H : **Hidden Layers**. Number of sequential ResBlocks inside a Context Net.

- **K : Kernel Size.** Spatial dimension of the convolutional filter.
- **D : Block Depth.** Number of consecutive convolutional layers inside a single ResBlock.
- **B : Batch Size.** Number of independent samples processed simultaneously.
- **F_{elem} : Element-wise FLOPs.** The exact number of floating-point operations required per spatial element for the affine transformation and Jacobian accumulation.

A standard Multiply-Accumulate (MAC) operation equates to **2 FLOPs** (1 multiplication + 1 addition). For a 2D convolution (X input channels, Y output channels, $K \times K$ kernel, $L \times L$ grid), the cost is $2 \cdot X \cdot Y \cdot K^2 \cdot L^2$.

2.1. Context Network ($FLOPs_{\text{net}}$)

Base Formula for 2D Convolution FLOPs

A single Multiply-Accumulate (MAC) operation counts as 2 FLOPs (one multiplication, one addition). For any standard 2D convolution with X input channels, Y output channels, a spatial resolution of $L \times L$, and a kernel size of $K \times K$, the theoretical FLOPs calculation is:

$$\text{FLOPs} = 2 \cdot X \cdot Y \cdot K^2 \cdot L^2$$

Step 1: Input Layer (1×1 Conv)

The first block in the Context Net, it takes the input field (C_{in} channels) and maps it to the hidden dimension (C channels) using a 1×1 kernel.

- **Channels:** $C_{\text{in}} \rightarrow C$
- **Kernel Area:** 1^2

$$\text{FLOPs}_{\text{in}} = 2 \cdot C_{\text{in}} \cdot C \cdot 1^2 \cdot L^2 = 2 \cdot C_{\text{in}} \cdot C \cdot L^2$$

Step 2: Hidden Layers (ResBlocks)

This section requires breaking down the nested repetitions shown in the diagram.

- **2a. Single Standard Conv2d:**

Inside the ResBlock, a standard convolution maintains the hidden channel dimension ($C \rightarrow C$) using a $K \times K$ kernel.

$$\text{FLOPs}_{\text{conv}} = 2 \cdot C \cdot C \cdot K^2 \cdot L^2 = 2 \cdot C^2 \cdot K^2 \cdot L^2$$

- **2b. Single ResBlock:**

The diagram notes that this standard Conv2d is "Repeated D times per block". (*Note: The element-wise residual addition $\oplus$ is omitted from this formula as its FLOP cost is infinitesimally small compared to matrix convolutions.*)

$$\text{FLOPs}_{\text{block}} = D \cdot (2 \cdot C^2 \cdot K^2 \cdot L^2)$$

- **2c. All ResBlocks:**

The entire ResBlock structure is "Repeated H times".

$$\text{FLOPs}_{\text{hidden}} = H \cdot \text{FLOPs}_{\text{block}} = H \cdot D \cdot 2 \cdot C^2 \cdot K^2 \cdot L^2$$

Step 3: Output Layer (1×1 Conv)

The final block in the Context Net reduces the hidden dimension (C channels) to the required affine parameters (C_{out} channels) using a 1×1 kernel.

- **Channels:** $C \rightarrow C_{\text{out}}$
- **Kernel Area:** 1^2

$$\text{FLOPs}_{\text{out}} = 2 \cdot C \cdot C_{\text{out}} \cdot 1^2 \cdot L^2 = 2 \cdot C \cdot C_{\text{out}} \cdot L^2$$

Step 4: Total Context Net Aggregation

To get the total computational cost of the Context Network, we sum the FLOPs from the three sequential stages:

$$\text{FLOPs}_{\text{net}} = \text{FLOPs}_{\text{in}} + \text{FLOPs}_{\text{hidden}} + \text{FLOPs}_{\text{out}}$$

$$\text{FLOPs}_{\text{net}} = (2 \cdot C_{\text{in}} \cdot C \cdot L^2) + (H \cdot D \cdot 2 \cdot C^2 \cdot K^2 \cdot L^2) + (2 \cdot C \cdot C_{\text{out}} \cdot L^2)$$

Step 5: Factoring the Equation

To arrive at target formula, we factor out the common mathematical terms shared across all three stages, which are $2 \cdot L^2 \cdot C$:

- From the Input Layer, we are left with: C_{in}
- From the Hidden Layers, we are left with: $H \cdot D \cdot C \cdot K^2$
- From the Output Layer, we are left with: C_{out}

Extracting these yields the final, simplified formula:

$$\text{FLOPs}_{\text{net}} = 2 \cdot L^2 \cdot C \cdot (C_{\text{in}} + C_{\text{out}} + H \cdot D \cdot C \cdot K^2)$$

2.2. Affine & Jacobian Operations ($\text{FLOPs}_{\text{affine}}$)

The element-wise mathematical functions applied to the unmasked grid:

$$\text{FLOPs}_{\text{affine}} = F_{\text{elem}} \cdot C_{\text{in}} \cdot L^2$$

2.3. Total FLOPs Formulas

For a **single sample** passing through all M layers:

$$\text{FLOPs}_{\text{sample}} = M \cdot (\text{FLOPs}_{\text{net}} + \text{FLOPs}_{\text{affine}})$$

For a **complete forward pass** with a batch size of B :

$$\text{FLOPs}_{\text{forward}} = B \cdot \text{FLOPs}_{\text{sample}}$$

3. Concrete FLOPs Calculation (Based on CONFIG)

We now apply the specific parameters from the provided configuration to calculate the exact theoretical FLOPs for one forward pass.

3.1. Parameter Mapping

Based on the `CONFIG` dictionary and scalar field theory:

- $L = 14$
- $M = 12(\text{cnn_coupling_layers})$
- $C_{\text{in}} = 1$ (Standard scalar field)
- $C_{\text{out}} = 2$ (Affine parameters s and t)
- $C = 256(\text{hidden_channels})$
- $H = 4(\text{hidden_layers})$
- $K = 3(\text{kernel_size})$
- $D = 3(\text{branch_depth})$
- $B = 1(\text{batch_size})$

3.2. Deriving the Element-wise Constant (F_{elem})

1. Affine Transformation (Field Update):

$$\phi'(x) = \phi_{\text{frozen}}(x) + m_u(x) \cdot [\phi(x) \cdot \exp(s(x)) + t(x)]$$

2. Jacobian Accumulation (Log-Determinant):

$$\log \det \left| \frac{\partial \phi'}{\partial \phi} \right| = \sum_x m_u(x) \cdot s(x)$$

For the unmasked spatial grid, the explicit operations per element are:

1. `exp(s)` $\rightarrow$ 1 FLOP
2. `phi * exp(s)` $\rightarrow$ 1 FLOP
3. `... + t` $\rightarrow$ 1 FLOP
4. `update_mask * (...)` $\rightarrow$ 1 FLOP
5. `phi_frozen + phi_updated` $\rightarrow$ 1 FLOP
6. `update_mask * s` $\rightarrow$ 1 FLOP
7. `sum(...)` $\rightarrow$ 1 FLOP (amortized addition across the grid)

Total exact element-wise FLOPs: $F_{\text{elem}} = 7$.

3.3. Step-by-Step Evaluation

Step A: Core Convolutional Arithmetic ($H \cdot D \cdot C \cdot K^2$)

$$4 \cdot 3 \cdot 256 \cdot 3^2 = 12 \cdot 256 \cdot 9 = 27648$$

Step B: Single Context Network ($FLOPs_{\text{net}}$)

$$\text{Summing the channels: } C_{\text{in}} + C_{\text{out}} + 27,648 = 1 + 2 + 27,648 = 27,651$$

$$\text{Multiplier: } 2 \cdot L^2 \cdot C = 2 \cdot 14^2 \cdot 256 = 2 \cdot 196 \cdot 256 = 100,352$$

$$FLOPs_{\text{net}} = 100,352 \cdot 27,651 = 2,774,833,152 \text{ FLOPs}$$

Step C: Single Affine & Jacobian Update ($FLOPs_{\text{affine}}$)

$$FLOPs_{\text{affine}} = 7 \cdot 1 \cdot 14^2 = 7 \cdot 196 = 1,372 \text{ FLOPs}$$

Step D: Complete Single Sample ($FLOPs_{\text{sample}}$)

Calculations for one Coupling Layer:

$$2,744,833,152 + 1,372 = 2,774,834,524 \text{ FLOPs}$$

Total for all $M = 12$ Layers:

$$FLOPs_{\text{sample}} = 12 \cdot 2,774,834,524 = 33,298,014,288 \text{ FLOPs}$$

Step E: Total Forward Pass ($FLOPs_{\text{forward}}$)

Factoring in the batch size ($B = 1$):

$$FLOPs_{\text{forward}} = 1 \cdot 33,298,014,288$$

$$FLOPs_{\text{forward}} = 33,298,014,288 \text{ FLOPs}$$

4. Analysis results using ptflops

Computational complexity: 16.68 GMac

Number of parameters: 85.06 M

Total FLOPs (1 MAC $\approx$ 2 FLOPs): 33.36 GFLOPs

5. Autocorrelation Time

The integral autocorrelation time (τ_{int}) of the physical quantity is approximately 2.1533.

The integral autocorrelation time prediction based on the MH rejection rate is 1.6944.

